



RUPA RAJA'S

PCMR E.M. SCHOOL

The Next Generation School

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CLASS - 10

CLICK
ME



Name of the Student: _____

Sec : Challengers

Sub: PHYSICAL SCIENCE

Time: 3 hours

Marks : 100

GRAND TEST 2

I. Answer the following questions.

8 x 1 = 8m

1. Why is photosynthesis considered an endothermic reaction?
2. Give any one example for neutralisation reaction?
3. Write the suffix and formula of functional group for Aldehyde.
4. Propose a method to extract a highly reactive metal from its Ore?
5. Write the formula of first member of the homologous series to which C_5H_{10} belongs.
6. Which of the following material cannot be used to make a lens?

a. Water

b. Glass

c. Plastic

d. Clay

7. Draw a symbol of electric resistance and Battery.
8. The radius of curvature of a spherical mirror is given as 20cm then determine its focal length.

II. Answer the following.

3 x 2 = 6m

9. Why does distilled water not conduct electricity, whereas rain water does?
10. Predict and write why copper and aluminium wires are usually employed for electricity transmission.

11.

Refractive Index	1.0003	1.31	1.33	1.50
Material medium	Air	Ice	Water	Benzene

A. Which medium is denser than water (or) ice?

B. Determine the speed of light in Benzene.

III. Answer the following questions.

3 x 4 = 12m

12. A. Draw the diagram which shows that acid solution in water conducts electricity.

(Or)

B. Draw the Diagrams for convex lens when the object is kept.

i. at infinity

iii. between $2f$ & f

ii. at f

iv. between p & f

13. Write any uses of metals and non-metals.

14. Draw a ray Diagrams for concave mirror when object kept.

i. At C

iii. Between C & F

ii. Between C & F

iv. At F

IV. Answer the following questions.

3 x 8 = 20m

15. A. Sai is not able to see the letters clearly far from her. Identify the eye defect she has been suffering from and how can you rectify it explain?

B. Derive the expression for resultant resistance when three resistors are connected in parallel.

16. A. Write the balanced chemical equations for the following reactions.

a. Calcium hydroxide + Carbon di Oxide $\rightarrow$ Calcium Carbonate + water.

b. Zinc + Silver nitrate $\rightarrow$ silver + zinc nitrate.

c. Aluminium + Copper chloride $\rightarrow$ Aluminium chloride + Copper.

d. Barium chloride + potassium sulphate $\rightarrow$ barium sulphate + potassium chloride.

(Or)

B. Complete the following table.

Functional group	Suffix	Formula of functional group	Example
Alcohol			CH ₃ OH
Aldehyde	- al		
		-Co-	CH ₃ COCH ₃
	- oicacid		CH ₃ COOH

17. A. Write an activity to show the water of crystallisation in salts?

(Or)

B. Describe an activity to show that the magnetic field produced around a current carrying conductor. Which deflects the compass needle?